CS1010: Discrete Mathematics for Computer Science

(Exam-2. Total: 30 marks.)
(Duration: 45 minutes. Date: 12 Oct 2026)

Instructions: ★ You may not get time to answer all the questions unless you have prepared really well. The exam is designed like that. ★ If your mobile phone is found with you during the exam, you will lose one grade. ★ Anybody found copying will get an F grade for the course straight away. ★ It is a no-break exam. You cannot take a break in between. The exam is only for 45 minutes. If you want to go out of the exam hall, you will have to submit your answer paper. ★ You should sit far apart from each other. The halls are big enough. If we see two students sitting close to each other, both the students will lose one grade.

$3^{2k+2+1} + 2^{k+1+2}$

$3^{2k+1} \times 9 + 2^{k+2} \times 2$

Questions

1. Using the principle of mathematical induction prove that, for every integer $n \geq 0$, $(3^{2n+1} + 2^{n+2})$ is divisible by 7. [5 marks]

2. Let a_n be the number of binary strings of length n that contain neither 00 nor 111 as a substring. For $n \geq 1$, let x_n, y_n, z_n denote, respectively, the number of **such strings** that end in 0, end in exactly one terminal 1, and end in exactly two terminal 1s.

   (a) Derive recurrence relations for x_n, y_n, z_n, together with their initial conditions at $n = 1$. [3 marks]

   (b) Hence, show that

   $$a_n = a_{n-2} + a_{n-3} \qquad (n \geq 4).$$

   [2 mark]

3. A machine accepts tokens worth 2, 5, and 7 rupees. The order in which tokens are inserted is irrelevant. Let c_n be the number of ways to deposit exactly n rupees subject to all three restrictions below:

   - an even number of 2-rupee tokens is used (zero is allowed);
   - at least one 5-rupee token is used;
   - at most two 7-rupee tokens are used.

$3^{2k+1}(7+2) + 2^{k+2} \times 2$

1 3

$3^{2k+1} \times 7 + 3^{2k+1} \times 2 + 2^{k+2} \times 2$

Model this question using a generating function in x such that the coefficient of x^n would be equal to c_n. (No need to solve this generating function to find the coefficient) [5 marks]

4. A group of n friends go on a trip. Each friend spends an integer number of rupees during the trip. Show that it is always possible to select a nonempty group of friends such that the total amount spent by the selected friends is divisible by n. [4 mark]

5. Find the number of ways to place $2t + 1$ indistinguishable balls in 3 distinct bins such that any two bins together contain more balls than the remaining bin. (*Give a brief justification for your answer.*) [3 mark]

6. Suppose that X_1 and X_2 are independent Bernoulli trials, each with probability 1/2, and let $X_3 = (X_1 + X_2) \bmod 2$.

   (a) Show that X_1, X_2, X_3 are pairwise independent, but X_3 and $X_1 + X_2$ are not independent. [2 mark]

   (b) Compute $V(X_1 + X_2 + X_3)$ and $V(X_1) + V(X_2) + V(X_3)$, where, $V(X)$ denotes the variance of the random variable X. [2 mark]

7. Design a procedure to select one winner uniformly at random from 20 people using an unbiased coin, so that each person has probability $\frac{1}{20}$ of being selected. [4 mark]

ALL THE BEST

$P(X) = \frac{1}{20}$

$\frac{1}{20} = \left(\frac{1}{2}\right)^n \leq \frac{1}{20}$

$n = 5$

$a_n = a_{n-2} + 2a_{n-4}$

$a_n = a_n + a_{n-1} + 2a_{n-2}$

$a_{n-1} = a_{n-3} + 2a_{n-5}$

a_{n-2}

$a_n + 2a_{n-3} + a_{n-4} + 2a_{n-5}$